

NOW !!!

Let

$$F(t) = \int_a^b f(x, t) dx$$

$f(x, t)$ and $\frac{\partial f(x, t)}{\partial t} = f_t(x, t)$ are continuous
at $a \leq x \leq b$
 $c \leq t \leq d$

Δt is an increment on t .

$$F(t + \Delta t) = \int_a^b f(x, t + \Delta t) dx$$

remember mean value
theorem for derivative
 $f(b) - f(a) = f'(c)(b - a)$
 $c \in (a, b)$

$$F(t + \Delta t) - F(t) = \int_a^b [f(x, t + \Delta t) - f(x, t)] dx$$

$(t + \Delta t - t) = \Delta t$

$$= \int_0^b f_t(x, t + \theta \Delta t) \Delta t \, dx \quad \boxed{0 < \theta < 1}$$

$$\frac{F(t + \Delta t) - F(t)}{\Delta t} = \int_a^b f_t(x, t + \theta \Delta t) \, dx$$

$$= \int_a^b f_t(x, t) + \int_a^b [f_t(x, t + \theta \Delta t) - f_t(x, t)] \, dx$$

for $\varepsilon > 0$; ^{since} $f_t(x, t)$ is continuous, there is a $\delta > 0$

$$|f_t(x, t + \theta \Delta t) - f_t(x, t)| < \varepsilon$$

since $|\Delta t| \leq \delta$

$$\begin{aligned} & \left| \frac{F(t+\Delta t) - F(t)}{\Delta t} - \int_a^b f_t(x, t) dx \right| \leq \\ & \leq \int_a^b |f_t(x, t+\theta\Delta t) - f_t(x, t)| dx \leq \epsilon(b-a) \end{aligned}$$

Thus

$$\lim_{\Delta t \rightarrow 0} \frac{F(t+\Delta t) - F(t)}{\Delta t} = \int_a^b f_t(x, t) dx$$

$$\frac{dF(t)}{dt} = \frac{d}{dt} \int_a^b f(x, t) dx = \int_a^b \frac{\partial f(x, t)}{\partial t} dx$$

$$= \int_a^b f_t(x, t) dx$$

int. bounds (limits) are constant.

NOW !!!

$$F(t) = \int_{a(t)}^{b(t)} f(x, t) dx$$

$a(t), b(t)$ are continuous relative to t .

You know

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + \frac{\partial F}{\partial b} \cdot \frac{db}{dt} + \frac{\partial F}{\partial a} \cdot \frac{da}{dt}$$

$$F(t) = \int_a^{b(t)} f(x, t) dx \Rightarrow \frac{\partial F}{\partial b} = f(b(t), t)$$

$$F(t) = \int_{a(t)}^b f(x, t) dx = - \int_b^{a(t)} f(x, t) dx \Rightarrow \frac{\partial F}{\partial a} = -f(a(t), t)$$

$$\rightarrow F'(t) = \int_{a(t)}^{b(t)} \frac{\partial f(x, t)}{\partial t} dx + f(b(t), t) \cdot b'(t) - f(a(t), t) \cdot a'(t)$$

Leibniz Rule is
obtained.

Namely;

$$\frac{d}{dx} F(x) = \frac{d}{dx} \int_0^x f(t) dt = f(x)$$

$$\int_{u(x)}^{v(x)} f(t) dt = \int_{u(x)}^c f(t) dt + \int_c^{v(x)} f(t) dt$$

$$\frac{d}{dx} \left[\int_{u(x)}^{v(x)} f(t) dt \right] = f(v(x)) \cdot v'(x) - f(u(x)) \cdot u'(x)$$

Ex. $F(t) = \int_t^{t^2} \sin x^2 dx$

$$F'(t) = \frac{d}{dt} \int_t^{t^2} \sin x^2 dx$$

$$= \int_t^{t^2} 0 \cdot dx + 2t \cdot \sin(t^2)^2 - 1 \cdot \sin t^2$$

$$\rightarrow F'(t) = \int_{a(t)}^{b(t)} \frac{\partial f(x, t)}{\partial t} dx + f(b(t), t) \cdot b'(t) - f(a(t), t) \cdot a'(t)$$

$$\text{Ex. } y = \int_0^{\sqrt{x}} e^{t^2} dt \Rightarrow \frac{dy}{dx} = ?$$

$$y' = \frac{1}{2\sqrt{x}} \cdot e^{(\sqrt{x})^2} - 0 \cdot e^0$$

$$\text{Ex: } \lim_{x \rightarrow \pi} \frac{\int_{\pi}^x \frac{dt}{2 + \cos t}}{x - \pi} = \lim_{x \rightarrow \pi} \frac{\frac{1}{2 + \cos x} - 0}{1} = 1$$

$$F(x) = \int_{u(x)}^{v(x)} f(x, y) dy$$

$$F'(x) = \int_{u(x)}^{v(x)} \frac{\partial f}{\partial x} dy + v'(x) \cdot f(x, v(x)) - u'(x) \cdot f(x, u(x))$$

Ex:

$$f(x) = \int_{1-x}^{2-x} e^{t^2} dt$$

$$f'(1) = ?$$

$$\begin{aligned} f'(x) &= e^{(2-x)^2} \cdot (2-x)^1 - e^{(1-x)^2} \cdot (1-x)^1 \\ &= e^{(2-x)^2} \cdot (-1) - e^{(1-x)^2} \cdot (-1) \end{aligned}$$

$$f'(x) = e^{(1-x)^2} - e^{(2-x)^2}$$

$$f'(1) = e^0 - e^1 = 1 - e$$

Ex:

$$\lim_{x \rightarrow 0} \frac{\arctan x}{\int_0^{2x} e^{t^2} dt} = ? \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{\frac{1}{1+x^2}}{e^{(2x)^2} \cdot 2 - e^0 \cdot 0} = \frac{1}{2} //$$

$$F(t) = \int_{x=1}^{x=t} \frac{\sin(x \cdot t)}{x} dx = ?$$

$$F'(t) = ?$$

$$F'(t) = \int_1^t \frac{\cancel{x} \cdot \cos(x \cdot t)}{\cancel{x}} dx + \frac{\sin t^2}{t} \cdot 1 - \frac{\sin t}{1} \cdot 0$$

$$F'(t) = \frac{1}{t} \cdot \sin(x \cdot t) \Big|_1^t + \frac{\sin t^2}{t}$$

$$F'(t) = \frac{1}{t} (\sin t^2) - \frac{1}{t} \sin t + \frac{\sin t^2}{t}$$

Ex:

$$F(x) = \frac{1}{x} \int_1^x [2t - F'(t)] dt \Rightarrow F'(1) = ?$$

$$F'(x) = -\frac{1}{x^2} \int_1^x \cancel{[2t - F'(t)]} dt + \frac{1}{x} \cdot [1 \cdot (2x - F'(x)) - 0.]$$

$$F'(1) = \frac{1}{1} (2 - F'(1))$$

$$2F'(1) = 2$$

$$\boxed{F'(1) = 1}$$

Find $g'(x)$ if $g(x) = \int_{x^2}^{\ln x} \arctan t \, dt$

Let F be any antiderivative of $\arctan t$. By the Evaluation Theorem, the integral in the question equals

$$F(\ln x) - F(x^2).$$

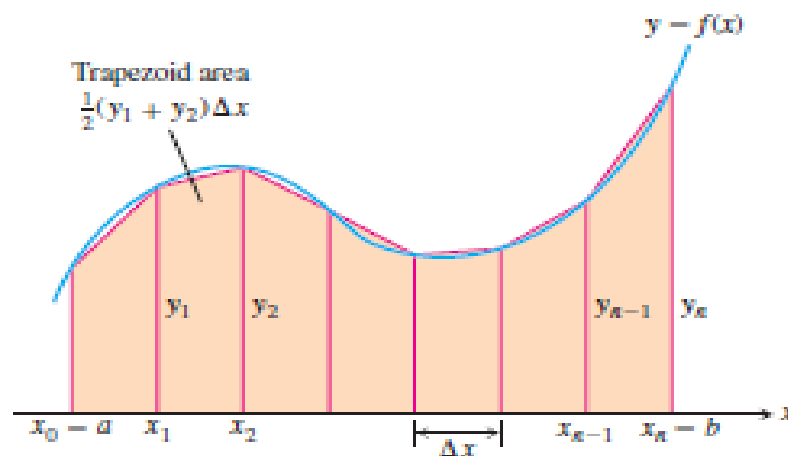
Taking the derivative using the chain rule (and remembering that, *by definition of antiderivative*, $F'(t) = \arctan t$), we get

$$\begin{aligned} g'(x) &= \left(\frac{1}{x}\right) \cdot F'(\ln x) - 2xF'(x^2) \\ &= \frac{\arctan(\ln x)}{x} - 2x \arctan(x^2). \end{aligned}$$

Numerical Integration

Trapezoidal Approximations

The Trapezoidal Rule for the value of a definite integral is based on approximating the region between a curve and the x -axis with trapezoids instead of rectangles, as in Figure



$$\Delta x = \frac{b - a}{n}.$$

$$\begin{aligned}
T &= \frac{1}{2}(y_0 + y_1)\Delta x + \frac{1}{2}(y_1 + y_2)\Delta x + \cdots \\
&\quad + \frac{1}{2}(y_{n-2} + y_{n-1})\Delta x + \frac{1}{2}(y_{n-1} + y_n)\Delta x \\
&= \Delta x \left(\frac{1}{2}y_0 + y_1 + y_2 + \cdots + y_{n-1} + \frac{1}{2}y_n \right) \\
&= \frac{\Delta x}{2}(y_0 + 2y_1 + 2y_2 + \cdots + 2y_{n-1} + y_n),
\end{aligned}$$

where

$$y_0 = f(a), \quad y_1 = f(x_1), \quad \dots, \quad y_{n-1} = f(x_{n-1}), \quad y_n = f(b).$$

The Trapezoidal Rule says: Use T to estimate the integral of f from a to b .

The Trapezoidal Rule

To approximate $\int_a^b f(x) dx$, use

$$T = \frac{\Delta x}{2} \left(y_0 + 2y_1 + 2y_2 + \cdots + 2y_{n-1} + y_n \right).$$

The y 's are the values of f at the partition points

$$x_0 = a, x_1 = a + \Delta x, x_2 = a + 2\Delta x, \dots, x_{n-1} = a + (n-1)\Delta x, x_n = b,$$

where $\Delta x = (b - a)/n$.

EXAMPLE Use the Trapezoidal Rule with $n = 4$ to estimate $\int_1^2 x^2 dx$. Compare the estimate with the exact value.

Solution Partition $[1, 2]$ into four subintervals of equal length $\Delta x = 1/4$. Then evaluate $y = x^2$ at each partition point.

Using these y values, $n = 4$, and $\Delta x = (2 - 1)/4 = 1/4$ in the Trapezoidal Rule, we have

$$\begin{aligned} T &= \frac{\Delta x}{2} \left(y_0 + 2y_1 + 2y_2 + 2y_3 + y_4 \right) \\ &= \frac{1}{8} \left(1 + 2\left(\frac{25}{16}\right) + 2\left(\frac{36}{16}\right) + 2\left(\frac{49}{16}\right) + 4 \right) \\ &= \frac{75}{32} = 2.34375. \end{aligned}$$

TABLE

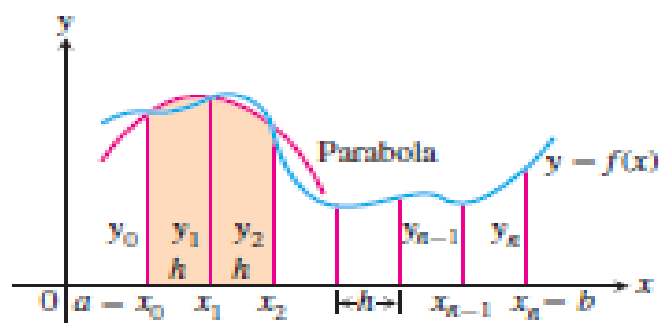
x	$y = x^2$
1	1
$\frac{5}{4}$	$\frac{25}{16}$
$\frac{6}{4}$	$\frac{36}{16}$
$\frac{7}{4}$	$\frac{49}{16}$
2	4

Since the parabola is concave *up*, the approximating segments lie above the curve, giving each trapezoid slightly more area than the corresponding strip under the curve. The exact value of the integral is

$$\int_1^2 x^2 dx = \left. \frac{x^3}{3} \right|_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}.$$

The T approximation overestimates the integral by about half a percent of its true value of $7/3$. The percentage error is $(2.34375 - 7/3)/(7/3) \approx 0.00446$, or 0.446%.

Simpson's Rule: Approximations Using Parabolas



Simpson's Rule

To approximate $\int_a^b f(x) dx$, use

$$S = \frac{\Delta x}{3} (y_0 + 4y_1 + 2y_2 + 4y_3 + \cdots + 2y_{n-2} + 4y_{n-1} + y_n).$$

The y 's are the values of f at the partition points

$$x_0 = a, x_1 = a + \Delta x, x_2 = a + 2\Delta x, \dots, x_{n-1} = a + (n-1)\Delta x, x_n = b.$$

The number n is even, and $\Delta x = (b - a)/n$.

EXAMPLE 1 Use Simpson's Rule with $n = 4$ to approximate $\int_0^2 5x^4 dx$.

Solution Partition $[0, 2]$ into four subintervals and evaluate $y = 5x^4$ at the partition points. Then apply Simpson's Rule with $n = 4$ and $\Delta x = 1/2$:

$$\begin{aligned} S &= \frac{\Delta x}{3} \left(y_0 + 4y_1 + 2y_2 + 4y_3 + y_4 \right) \\ &= \frac{1}{6} \left(0 + 4\left(\frac{5}{16}\right) + 2(5) + 4\left(\frac{405}{16}\right) + 80 \right) \\ &= 32\frac{1}{12}. \end{aligned}$$

TABLE 1

x	$y = 5x^4$
0	0
$\frac{1}{2}$	$\frac{5}{16}$
1	5
$\frac{3}{2}$	$\frac{405}{16}$
2	80

EXAMPLE Use the Trapezoidal Rule to approximate the integral $\int_1^2 (1/x) dx$.

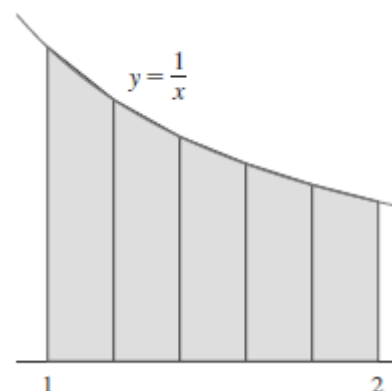
with $n = 5$ to

SOLUTION

With $n = 5$, $a = 1$, and $b = 2$, we have $\Delta x = (2 - 1)/5 = 0.2$, and so the Trapezoidal Rule gives

$$\begin{aligned}\int_1^2 \frac{1}{x} dx &\approx T_5 = \frac{0.2}{2} [f(1) + 2f(1.2) + 2f(1.4) + 2f(1.6) + 2f(1.8) + f(2)] \\ &= 0.1 \left(\frac{1}{1} + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2} \right) \\ &\approx 0.695635\end{aligned}$$

$$\int_1^2 \frac{1}{x} dx = \ln x \Big|_1^2 = \ln 2 = 0.693147 \dots$$



EXAMPLE Use Simpson's Rule with $n = 10$ to approximate $\int_1^2 (1/x) \, dx$.

SOLUTION Putting $f(x) = 1/x$, $n = 10$, and $\Delta x = 0.1$ in Simpson's Rule, we obtain

$$\begin{aligned}\int_1^2 \frac{1}{x} \, dx &\approx S_{10} \\&= \frac{\Delta x}{3} [f(1) + 4f(1.1) + 2f(1.2) + 4f(1.3) + \cdots + 2f(1.8) + 4f(1.9) + f(2)] \\&= \frac{0.1}{3} \left(\frac{1}{1} + \frac{4}{1.1} + \frac{2}{1.2} + \frac{4}{1.3} + \frac{2}{1.4} + \frac{4}{1.5} + \frac{2}{1.6} + \frac{4}{1.7} + \frac{2}{1.8} + \frac{4}{1.9} + \frac{1}{2} \right) \\&\approx 0.693150\end{aligned}$$